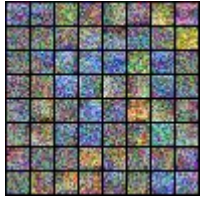


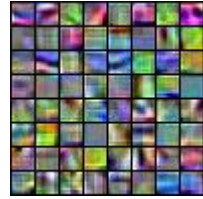
forward feature learning

backward feature correction



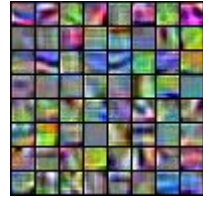
epoch 1

...



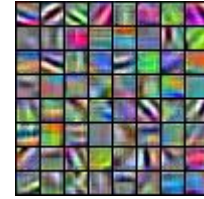
epoch 60

...



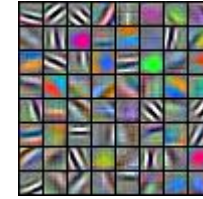
epoch 80

...



epoch 100

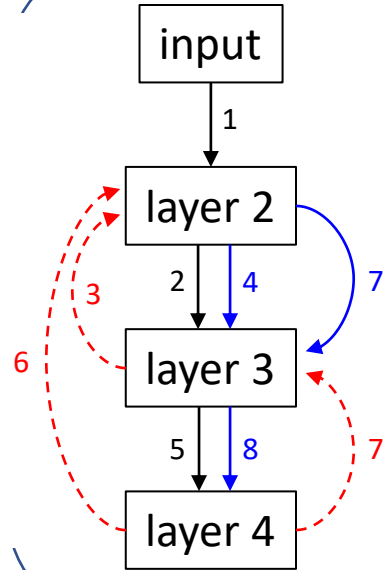
...



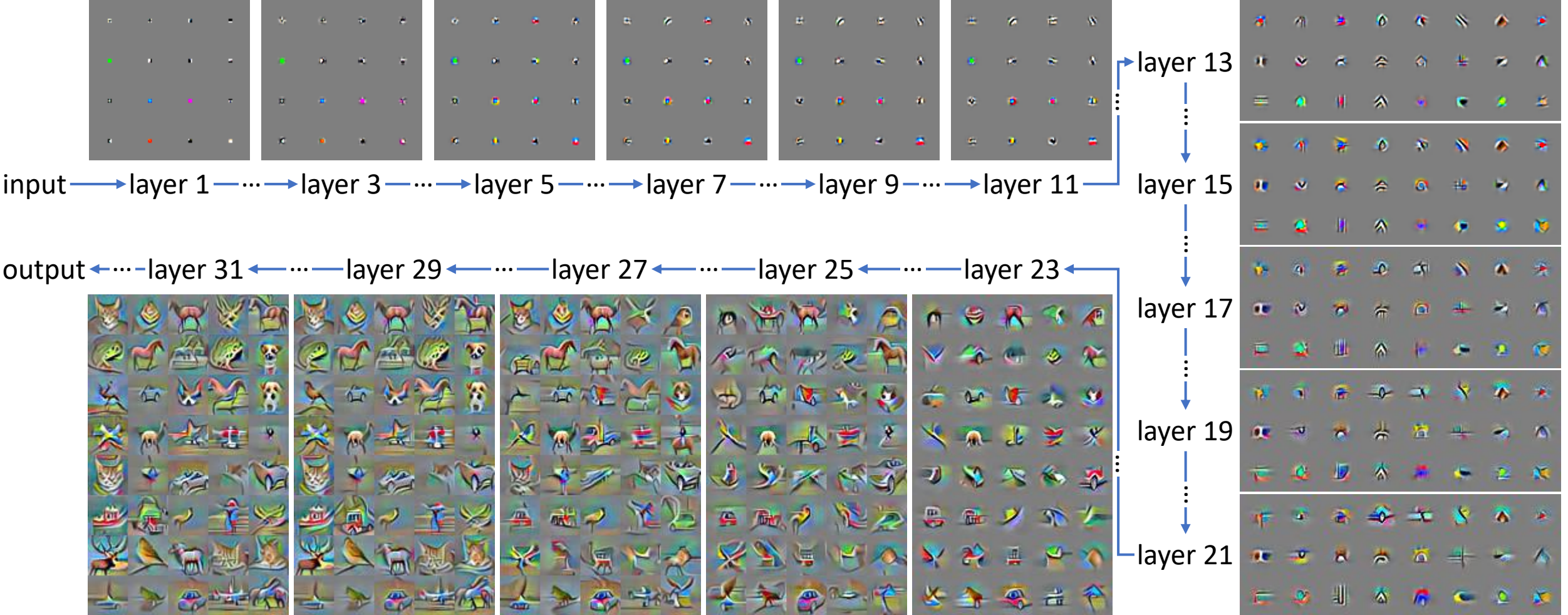
epoch 200

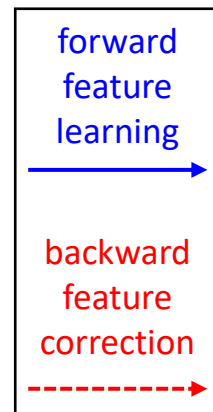
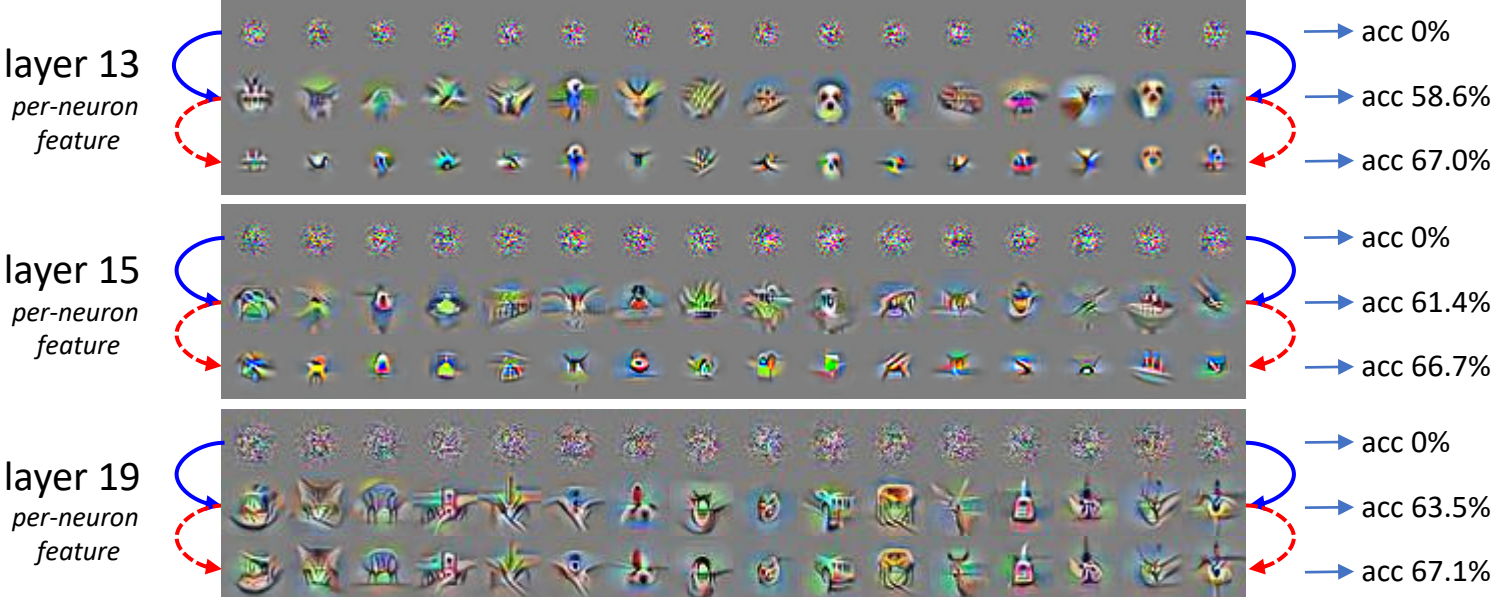
1st layer features no longer improve through training only 1st layer

1st layer features improve again once we start to also train higher-level layers

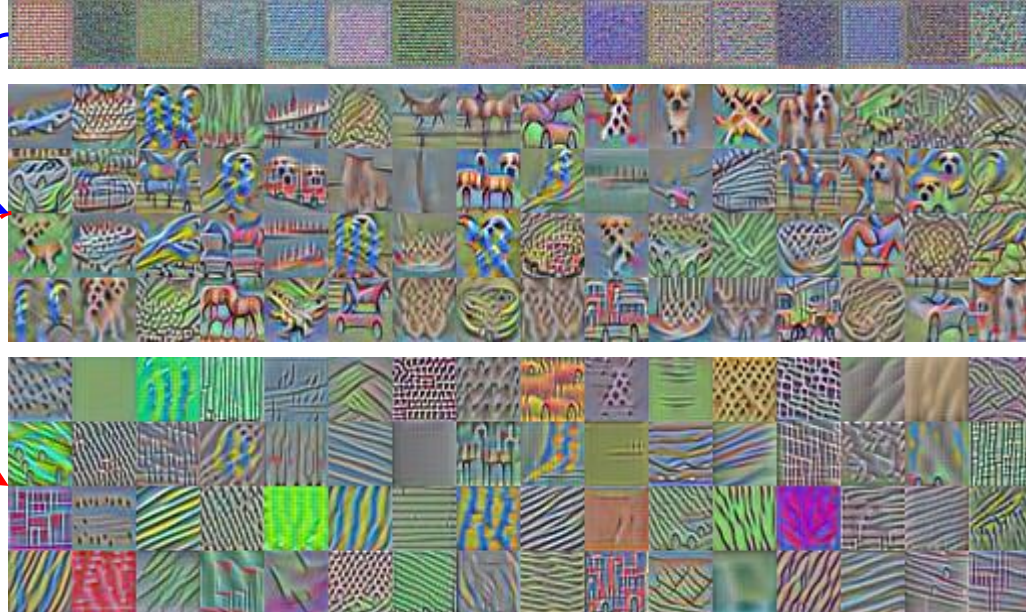


- 1) layer 2 begins to learn
- 2) layer 3 begins to learn using features given by layer 2
- 3) layer 3 helps layer 2 correct its features by reducing over-fitting
- 4) layer 3 improves since layer 2 now feeds in better features to layer 3
- 5) layer 4 begins to learn
- 6) layer 4 helps layer 2 correct its features by reducing over-fitting
- 7) layer 3 improves since
 - a) layer 4 helps layer 3 correct its features by reducing over-fitting, and
 - b) layer 2 now feeds in better features to layer 3
- 8) layer 4 improves since layer 3 now feeds in better features





layer 13
per-channel
feature



→ acc 0.0% (random init)

forward feature learning

→ acc 58.6% (if only train ≤ 13 layers)

backward feature correction

→ acc 67.0% (if train all layers together)

	CIFAR-100 accuracy	$\ell = 1$	$\ell = 3$	$\ell = 5$	$\ell = 7$	$\ell = 9$	$\ell = 11$	$\ell = 13$	$\ell = 15$	$\ell = 17$	$\ell = 19$	$\ell = 21$	
ensemble	train only $\leq \ell$	16.0%	43.1%	61.5%	67.9%	70.7%	71.5%	75.9%	78.6%	79.8%	80.6%	80.9%	→ no BFC
	fix $\leq \ell$, train the rest	83.1%	78.9%	78.4%	76.9%	75.9%	75.6%	77.6%	79.6%	80.5%	81.0%	81.2%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	83.4%	81.5%	79.8%	78.4%	77.4%	78.7%	80.3%	80.7%	81.0%	81.3%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	83.1%	81.9%	81.2%	80.1%	80.8%	82.6%	82.2%	81.0%	81.3%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	83.2%	82.3%	82.0%	81.8%	82.4%	82.7%	82.0%	81.8%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	83.4%	82.2%	82.4%	83.1%	83.2%	82.7%	81.9%	→ BFC for 8 layers
	train all the layers	83.2%	83.2%	83.0%	82.9%	82.8%	83.0%	83.1%	83.1%	82.9%	83.2%	83.0%	→ full BFC
	average weight correlations (for “train $\leq \ell$ ” vs “rand init”)	0.131	0.081	0.070	0.054	0.051	0.037	0.036	0.034	0.034	0.032	0.031	→ training neural nets is far from the NTK regime
single model	average weight correlations (for “train $\leq \ell$ ” vs “train all”)	0.927	0.956	0.966	0.958	0.965	0.960	0.950	0.968	0.967	0.959	0.948	→ correlation between with vs. without BFC
	train only $\leq \ell$	16.0%	41.1%	58.5%	64.0%	67.3%	68.0%	72.0%	74.8%	76.0%	76.1%	76.9%	→ no BFC
	fix $\leq \ell$, train the rest	79.4%	72.9%	72.7%	70.6%	69.8%	69.9%	72.1%	74.2%	75.2%	75.2%	76.1%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	79.6%	77.5%	74.5%	72.3%	71.5%	72.8%	74.7%	75.0%	75.1%	75.8%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	79.9%	78.1%	76.2%	74.8%	75.2%	77.3%	76.9%	75.8%	75.9%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	80.0%	78.6%	77.1%	76.8%	78.0%	78.9%	77.4%	76.8%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	79.9%	78.4%	78.3%	78.7%	79.2%	79.0%	77.6%	→ BFC for 8 layers
	train all the layers	79.7%	79.9%	79.6%	79.4%	79.3%	79.4%	79.4%	79.1%	79.0%	79.3%	79.1%	→ full BFC

	CIFAR-10 accuracy	$\ell = 1$	$\ell = 3$	$\ell = 5$	$\ell = 7$	$\ell = 9$	$\ell = 11$	$\ell = 13$	$\ell = 15$	$\ell = 17$	$\ell = 19$	$\ell = 21$	
ensemble	train only $\leq \ell$	41.7%	73.5%	89.1%	92.2%	93.4%	93.9%	94.2%	95.1%	95.7%	95.9%	96.1%	→ no BFC
	fix $\leq \ell$, train the rest	96.5%	93.7%	91.7%	92.8%	93.7%	94.2%	94.3%	95.3%	95.9%	96.1%	96.3%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	96.5%	95.5%	93.4%	93.6%	94.1%	94.4%	95.2%	95.8%	95.9%	96.3%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	96.4%	96.1%	94.9%	94.5%	94.4%	95.5%	95.6%	95.8%	96.1%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	96.7%	96.1%	95.4%	95.0%	95.6%	96.0%	95.6%	95.9%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	96.6%	96.3%	95.9%	96.2%	96.3%	96.1%	95.7%	→ BFC for 8 layers
	train all the layers	96.5%	96.6%	96.5%	96.5%	96.5%	96.4%	96.2%	96.3%	96.4%	96.5%	96.6%	→ full BFC
	average weight correlations (for “train $\leq \ell$ ” vs “rand init”)	0.276	0.083	0.068	0.060	0.053	0.043	0.043	0.040	0.036	0.037	0.031	→ training neural nets is far from the NTK regime
single model	average weight correlations (for “train $\leq \ell$ ” vs “train all”)	0.905	0.977	0.975	0.972	0.966	0.965	0.964	0.966	0.959	0.957	0.939	→ correlation between with vs. without BFC
	train only $\leq \ell$	41.4%	71.2%	86.5%	89.9%	91.3%	91.8%	92.3%	93.9%	94.5%	94.7%	95.0%	→ no BFC
	fix $\leq \ell$, train the rest	95.6%	90.8%	89.1%	90.4%	91.5%	92.0%	92.5%	93.9%	94.6%	94.8%	95.0%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	95.7%	93.7%	90.9%	91.3%	91.8%	92.3%	93.7%	94.4%	94.6%	94.9%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	95.7%	94.7%	93.0%	92.2%	92.4%	93.9%	94.1%	94.2%	94.6%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	95.8%	94.9%	93.8%	93.2%	94.2%	94.6%	94.1%	94.2%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	95.8%	95.1%	94.5%	94.7%	95.0%	94.8%	94.1%	→ BFC for 8 layers
	train all the layers	95.8%	95.6%	95.6%	95.8%	95.6%	95.5%	95.4%	95.7%	95.9%	95.8%	95.9%	→ full BFC

single model	CIFAR-10, L2 adversarial	$\ell = 1$	$\ell = 3$	$\ell = 5$	$\ell = 7$	$\ell = 9$	$\ell = 11$	$\ell = 13$	$\ell = 15$	$\ell = 17$	$\ell = 19$	$\ell = 21$	
	train only $\leq \ell$	23.5%	36.2%	47.0%	52.6%	55.9%	57.3%	58.7%	61.9%	63.3%	63.9%	64.4%	→ no BFC
	fix $\leq \ell$, train the rest	67.2%	63.4%	64.3%	62.5%	58.3%	57.8%	59.3%	61.8%	62.6%	63.7%	63.7%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	67.3%	65.7%	63.7%	62.3%	60.1%	59.9%	61.6%	62.7%	63.7%	64.1%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	67.0%	65.9%	64.9%	63.1%	62.1%	63.5%	63.3%	63.5%	64.0%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	67.0%	65.8%	64.7%	64.5%	65.3%	65.6%	64.7%	63.9%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	67.0%	66.5%	65.8%	66.0%	66.3%	66.4%	65.1%	→ BFC for 8 layers
	train all the layers	66.7%	67.0%	66.9%	66.8%	66.7%	66.7%	66.7%	66.5%	67.0%	66.0%	65.5%	→ full BFC
	average weight correlations (for “train $\leq \ell$ ” vs “rand init”)	0.354	0.134	0.108	0.089	0.074	0.065	0.059	0.050	0.045	0.045	0.042	→ training neural nets is far from the NTK regime
	average weight correlations (for “train $\leq \ell$ ” vs “train all”)	0.798	0.868	0.877	0.864	0.911	0.890	0.862	0.860	0.861	0.834	0.802	→ correlation between with vs. without BFC

single model	CIFAR-10, L_{inf} adversarial	$\ell = 1$	$\ell = 3$	$\ell = 5$	$\ell = 7$	$\ell = 9$	$\ell = 11$	$\ell = 13$	$\ell = 15$	$\ell = 17$	$\ell = 19$	$\ell = 21$	
	train only $\leq \ell$	21.6%	28.4%	37.6%	42.8%	46.7%	48.1%	50.6%	54.1%	55.6%	56.5%	57.4%	→ no BFC
	fix $\leq \ell$, train the rest	60.6%	55.7%	56.7%	54.5%	52.6%	51.0%	52.0%	54.0%	55.5%	56.6%	57.5%	→ no BFC
	fix $\leq \ell - 2$, train the rest	-	61.3%	58.2%	55.9%	54.2%	53.3%	53.1%	55.6%	56.4%	56.5%	57.2%	→ BFC for 2 layers
	fix $\leq \ell - 4$, train the rest	-	-	61.0%	58.6%	57.0%	56.3%	55.7%	57.5%	57.8%	57.3%	57.6%	→ BFC for 4 layers
	fix $\leq \ell - 6$, train the rest	-	-	-	61.4%	59.9%	57.6%	58.0%	59.1%	59.5%	58.7%	58.3%	→ BFC for 6 layers
	fix $\leq \ell - 8$, train the rest	-	-	-	-	61.1%	59.3%	58.6%	59.4%	60.0%	60.4%	60.0%	→ BFC for 8 layers
	train all the layers	60.6%	60.5%	61.1%	61.3%	60.7%	60.1%	60.5%	60.5%	60.2%	60.4%	60.8%	→ full BFC
	average weight correlations (for “train $\leq \ell$ ” vs “rand init”)	0.164	0.137	0.086	0.070	0.061	0.054	0.054	0.041	0.047	0.040	0.037	→ training neural nets is far from the NTK regime
	average weight correlations (for “train $\leq \ell$ ” vs “train all”)	0.622	0.783	0.845	0.835	0.898	0.883	0.847	0.867	0.849	0.942	0.904	→ correlation between with vs. without BFC

